Reg. No. | | | | | | | | | | | [16CS602/16IS663]

SIXTH SEMESTER B. TECH EXAMINATION, JUNE – 2019

PARALLEL AND DISTRIBUTED COMPUTING

Time: 3 Hours **Maximum Marks:70**

Instructions: 1. Answer any **FIVE** full questions choosing atleast two from each part.
2. Missing data may be suitably assumed.

PART - A

1. a. Define Parallel Computing. What are the main reasons for using Parallel Computing? **04**

   b. Discuss in detail about; **10**

   i. Multi-Core Architecture

   ii. Many Core Architecture

2. a. Define the following with reference to Parallel Computing **04**

   i. Granularity

   ii. Parallel Overhead

   iii. Embarrassingly Parallel

   iv. Observed Speedup

   b. Write Short notes on **06**

   i. Open MP Threads

   ii. POSSIX Threads

   c. List the Pros and Cons of Message Passing Model and Shared Memory Model. **04**

3. a. Briefly explain the steps to be followed for solving a problem using parallelism. **06**

   b. With neat sketch, explain the fixed function GPU Pipeline. **08**

4. a. Portray the four different memory regions accessed by a work item in OpenCL memory mode and compare those memory regions. **08**

   b. Write short notes on Half Data type and describe the different types of Address Space Qualifiers in OpenCL in OpenCL. **06**

PART - B

5. a. Discuss in detail about **08**

   i. Workstation Sever Model

   ii. Processor Pool Model

   b. Describe the various types of Buffering in Message Passing Model. **04**

   c. What are the three primary reasons for failure in Message Passing Model? **02**

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6. a. Draw and explain the message format of RPC Call Messages and Reply Messages. **06**

   b. What are the necessary conditions for Deadlock? Explain Deadlock Avoidance and Deadlock Prevention detail. **08**

7. a. Briefly discuss about the taxonomy of Load Balancing algorithms. **07**

   b. Explain in detail about the 3 models for Organizing Threads. **07**

8. a. Elucidate the 3 different Cache Locations in Distributed File System with clear diagram. **07**

   b. Explain the Sateful and Stateless File Servers with respect to Fault Tolerance. **07**